How Much of Platonic Convergence is Semantic? Measuring Alignment in the Jacobian Subspace

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Abstract

The Platonic Representation Hypothesis [13] asserts that neural networks converge towards a shared representation of reality as they become more competent. Current evidence for this hypothesis comes from full activation space alignment, which mixes semantic content with superficial regularities like document length and structure. We instead measure alignment in the Jacobian space (J-space), the privileged subspace induced by the Jacobian lens (J-lens) [11], and find that restricting the comparison to this subspace makes alignment more sensitive to semantic content, though a control attributes only part of this effect to the J-space itself.

Measuring alignment in the J-space, we still find that models converge: the text–text Spearman correlation between competence and alignment is $\rho = +0.56$, $p = 0.0162$, and the text–vision correlation is $\rho = +0.97$, $p < 10^{-4}$. Within language, though, the J-space converges much more slowly than the full activation space, so part of the apparent link between competence and convergence is carried by superficial structure. Consistent with this, the gap between J-space and full convergence disappears across modalities, where no such structure is shared.

This convergence is also legible: using the J-lens at matched layer depth, independently trained models name 5.1 of 25 shared vocabulary words against 2.0 for the plain logit lens. J-lens overlap rises with competence ($\rho = +0.74$, $p = 0.0020$). Overall, full-space alignment is partially confounded by superficial structure, and restricting alignment comparison to the J-space can help isolate semantically meaningful convergence. Code available at `https://anonymous.4open.science/r/jspace-convergence-anon/`.

1 Introduction

The Platonic Representation Hypothesis [13] asserts that "Neural networks, trained with different objectives on different data and modalities, are converging to a shared statistical model of reality in their representation spaces." We use two terms throughout this paper: a model pair's *alignment* is how similar the representations of the two models are, and *convergence* is the correlation between alignment and competence, across model pairs. The Platonic hypothesis claims this correlation is positive.

To our knowledge, alignment has thus far been computed over full representational spaces.[1] We show this method conflates semantic content with confounding surface-level similarities like structure, length, and formatting.

[1] We say "full activation space" rather than "full representational space", since every quantity we compare is a mean-pooled residual-stream vector at a given layer.

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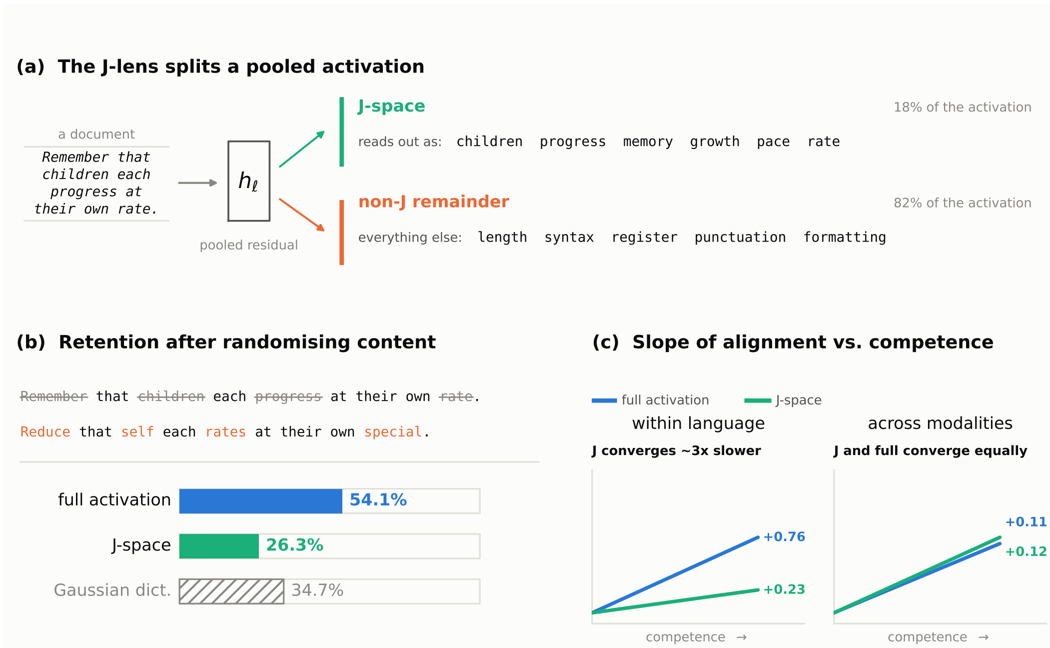


Figure 1: **Measuring Platonic convergence in the J-space.** **(a)** The J-lens splits a pooled activation into a J-space component and a non-J remainder. The document shown above is the example found in Section 3.1. The list of words is illustrative. **(b)** Randomising content words retains 26.3% of J-space alignment and 54.1% of full-activation alignment. Under the same measurement with a random Gaussian dictionary rather than a J-lens dictionary, alignment retains 34.7% (Table 1). **(c)** Within language, J-space alignment rises with competence at approximately one third the rate of full alignment. Across modalities, J-space alignment and full activation alignment converge at around equal rates (Figure 2).

Recently, Gurnee et al. [11] introduced a new interpretability technique, the Jacobian lens (J-lens), which recovers the concepts a language model is poised to verbalise at a given layer. The J-lens induces the J-space, a privileged subspace of internal representations available for report and reasoning. It is semantically significant, acting as a "global workspace" for the model, and privileged, capturing a median of only 6–7% of a concept vector's variance [11].

Thus, we hypothesise that restricting the comparison to J-spaces will make alignment more sensitive to semantic content than the full-activation comparison, in the sense that J-space alignment will drop more sharply when semantic content is removed (Figure 1).

Motivating experiment We compute alignment using "mutual κ-nearest-neighbour overlap" between pairs of models. Other metrics are reported in Appendix D. We do this once on an ordinary dataset of text documents and once on the same documents with content words randomised but sentence structure preserved. We find that over half of the full alignment is retained, compared with just over a quarter of J-space alignment. A random-dictionary control separates how much of this gap is attributable to the J-lens itself as opposed to the sparse coding effect of decomposition (Section 4.1).

We use the J-space to re-examine the Platonic Representation Hypothesis's central claim, and find that models do indeed converge under this subspace-restricted comparison.

Our three experiments compare 1) J-space against full alignment for text models (including the content-ablated motivating experiment); 2) text models against vision encoders, which have no J-space of their own; and 3) the J-lens readouts of text models, where outputs are vocabulary tokens and no additional alignment machinery is needed. For each case, we calculate convergence over model pairs. The third experiment also compares J-lens overlap with logit lens overlap [20] as a control.

Related work Representational similarity has been measured in many ways: RSA compares dissimilarity matrices [16], SVCCA compares subspaces through canonical correlation [25], and centred kernel alignment (CKA) compares item–item kernels, with invariance to orthogonal transformation and isotropic scaling [15]. Huh et al. [13] measure convergence using mutual κ-nearest-neighbour overlap [22] and CKNNA. Some work [25, 29] does restrict attention to a subspace, but always by introducing a new measure of alignment.

We instead vary only the *subspace* the metric is computed in, holding the metric itself fixed. Appendix D repeats the comparison under seven further metrics.

Platonic convergence is also contested. Gröger et al. [10] suggest model alignment is inflated, and Bangachev et al. [1] partly attribute alignment to word frequency. It is well-established that alignment measures capture non-semantic structure. We instead examine how much of the competence–convergence relationship is semantic by comparing the J-spaces of models. To our knowledge, the J-space has not yet been used to examine convergence.

Contributions

1. We propose measuring alignment between models in the J-space rather than the full activation space, and we validate this subspace-restricted comparison as more sensitive to semantic content using an intervention. We find that randomising content words removes 74% of J-space alignment but only 46% of full-activation alignment, though only about one third of this gap can be directly attributed to the J-space rather than its sparse coding.
2. We find that part of the existing competence–convergence relationship rests on structure that is shared, but not semantic. Indeed, within language, J-space alignment rises with competence at roughly a third the rate of the full activation space, while across modalities (where no such shared structure exists) the two rates are indistinguishable.
3. We show that J-space convergence is meaningful and can be read as words using the J-lens. At matched relative layer depth, independently trained models name 5.1 of 25 shared vocabulary words against 2.0 for the plain logit lens. J-lens readouts also converge as competence increases, at least as closely as the neighbour-overlap measure.

2 Formal setup

We first define the alignment measure used in Experiments 1 and 2. Our third experiment needs no machinery beyond the J-lens itself, since the readouts are in a shared vocabulary space.

For the first two experiments there is no such shared output space. For instance, two models may have different residual widths or different modalities. We therefore adapt the method of Huh et al. [13], which measures alignment by kernels rather than the representations themselves. Between language models, we compare the kernels induced by the J-spaces of the models. Across modalities, we compare the kernel induced by the J-space of the text model to the kernel built from raw pooled patch features.

Let $A, B, \ldots$ denote language models with residual widths $r_A, r_B, \ldots$ and layer counts $L_A, L_B, \ldots$ that generally differ. We index layers by $\ell \in \{0, \ldots, L_A - 1\}$ and write $\lambda = \ell/(L_A - 1) \in [0, 1]$ for the corresponding *relative depth*.

We consider alignment over input items. For text–text comparisons, item i is a document and both models receive it. For text–vision comparisons, item i is an image–caption pair, with each encoder receiving its respective input. For each model, layer ℓ, and item i, we mean-pool the residual stream over the item's tokens or patches, giving $h_i^{A,(\ell)} \in \mathbb{R}^{r_A}$.

For each language model, we compute the fitted J-lens [11]:

$$J_\ell^A \;=\; \mathbb{E}_i\left[\,\mathbb{E}_{(s' \geq s)}\left[\frac{\partial h_{s',i}^{A,(\text{final})}}{\partial h_{s,i}^{A,(\ell)}}\right]\right] \;\in\; \mathbb{R}^{r_A \times r_A},$$

where s and s' index tokens with $s' \geq s$, since a token cannot affect its predecessors.

Let $W_U^A \in \mathbb{R}^{m_A \times r_A}$ be model A's unembedding matrix, mapping its final residual stream to m_A vocabulary logits. We preprocess W_U^A by folding in the gain w^A of model A's final normalisation layer and centring it over the m_A output coordinates:

$$\widetilde{W}_U^A \;=\; \left(I_{m_A} - \frac{1}{m_A}\mathbf{1}\mathbf{1}^\top\right) W_U^A \,\mathrm{diag}(w^A), \qquad \mathbf{1} \in \mathbb{R}^{m_A}.$$

The J-space dictionary for layer ℓ of model A, with rows $d^A_{\ell,v}$ that are directions in residual-stream space, is

$$D_\ell^A \;=\; \widetilde{W}_U^A J_\ell^A \;\in\; \mathbb{R}^{m_A \times r_A}.$$

2.1 Decomposing activations into the J-space and remainder

We write each pooled activation as a sparse non-negative combination of $k = 25$ J-space dictionary directions, which is how we split the activations into their J-space and non-J remainder. To select the subset $\mathcal{S}_i$ of dictionary coordinates with $|\mathcal{S}_i| = k$, we use a greedy orthogonal matching pursuit (OMP), refitting coefficients $\alpha_{i,v}$ with a projected-gradient non-negative least-squares (NNLS) step.

Denote by $\bar{d}^A_{\ell,v}$ the normalised unit directions, and write $h^{A,(\ell)}_{\mathrm{full},i} \equiv h^{A,(\ell)}_i$ for the full pooled activation. Then, we have

$$h^{A,(\ell)}_{J,i} = \sum_{v \in \mathcal{S}_i} \alpha_{i,v}\, \bar{d}^A_{\ell,v}, \qquad h^{A,(\ell)}_{\perp,i} = h^{A,(\ell)}_{\mathrm{full},i} - h^{A,(\ell)}_{J,i}.$$

We thus consider three components, written $h^{A,(\ell)}_{\omega,i}$ for $\omega \in \{\mathrm{full}, J, \perp\}$: the full activation $h^{A,(\ell)}_{\mathrm{full},i}$, the J-space component $h^{A,(\ell)}_{J,i}$, and the non-J remainder $h^{A,(\ell)}_{\perp,i}$. Our aim is to compare alignment computed from each.

2.2 Kernels

Each component is winsorised at the 95th percentile and normalised to unit magnitude, and we denote the preprocessed components by $\hat{h}^{A,(\ell)}_{\omega,i}$. The kernel for model A, layer ℓ, and component ω is the $n \times n$ matrix of cosine similarities between items,

$$K_\ell^{A,\omega}(i,j) = \left\langle \hat{h}^{A,(\ell)}_{\omega,i},\, \hat{h}^{A,(\ell)}_{\omega,j} \right\rangle.$$

The dictionaries of text models have one row per vocabulary token [3, 8], but a vision analogue would not be as easily interpretable: MAE [12] decodes to raw pixels, and DINOv2 [21] decodes to abstract prototypes. We thus apply the J-lens only to the caption side, building image kernels from raw pooled patch features, so on the image side, only $\omega = \mathrm{full}$ is defined. The image kernel $K_\ell^{\mathrm{img,\,full}}$ is formed by the same cosine-similarity expression from the preprocessed[2] mean-pooled patch activations.

2.3 Mutual κ-nearest-neighbour alignment

We compare two kernels through mutual κ-nearest-neighbour overlap [22]. Taking each item's κ nearest neighbours under each kernel (the κ largest off-diagonal entries of its row), we define $\mathrm{m\text{-}NN}(K_\ell^{A,\omega},\, K_{\ell'}^{B,\omega'})$ between layer ℓ of model A and layer ℓ' of model B as the average fraction of neighbours shared by the two kernels, over all n items.

For a text–text comparison, both sides carry the same component, $\omega' = \omega$. For a text–vision comparison, the caption side varies over $\omega \in \{\mathrm{full}, J, \perp\}$ while the image side is fixed at $\omega' = \mathrm{full}$.

Finally, we define the alignment between A and B for component ω as the mean over layer pairs $(\ell, \ell') \in \mathcal{B}_A \times \mathcal{B}_B$, where $\mathcal{B}_A = \{\, \ell : 0.35 \le \ell/(L_A - 1) \le 0.90 \,\}$ is the relative layer depth band:

$$\mathcal{A}^\omega(A,B) \;=\; \mathrm{mean}_{\ell,\ell'}\, \mathrm{m\text{-}NN}\big(K_\ell^{A,\omega},\, K_{\ell'}^{B,\omega'}\big).$$

[2] The image side is winsorised per coordinate at the 0.5th and 99.5th percentiles, then normalised to unit magnitude.

3 Experiment setup

3.1 Text–text alignment and motivating experiment

We compare the following 11 text models of varying size and architecture: Pythia-70M [2], GPT-2 [23], Gemma-3-270M [7], Qwen3.5-0.8B [27], Gemma-3-1B, Qwen3-1.7B [28], Qwen3.5-2B, Gemma-2-2B [6], Qwen3-4B, Qwen3.5-4B, and Gemma-3-4B.

For the first experiment, we use the prefitted J-lenses on Neuronpedia [17], and compute alignment across all $\binom{11}{2} = 55$ text–text pairs on 1,000 documents from pile-10k [19], a subset of the Pile [5]. We take the first 1,000 rows, truncate each to at most 1,500 characters, then to at most 300 tokens, and mean-pool over token positions. Any difference in tokeniser efficiency is quite marginal, from 3.79 to 4.19 characters per token. Measured over the corpus, the least efficient tokeniser still ingests 94.8% as many characters per document as the most efficient. Neighbourhoods are taken at $\kappa = 10$.

Following the procedure described in the previous section, we compute alignment for the J-space, the non-J remainder, and the full activation space over the original corpus. We then repeat the measurement on a content-ablated variant of the same 1,000 documents. We preserve the sentence structure, replacing each non-stopword alphabetic token (i.e. each content word) with another drawn from the same log-frequency bin. For example, “Remember that children each progress at their own rate.” becomes “Reduce that self each rates at their own special.” Comparing these two results isolates how much of each component’s alignment is semantic.

To assess how much of the retention drop comes from the sparse coding of the J-space, we run a Gaussian dictionary through the same content-ablation test and compute its retention rate.

Finally, we relate alignment to competence, measuring competence[3] using the HellaSwag benchmark [30]. Appendix B repeats the analysis with different competence axes.

3.2 Text–vision alignment

For our second experiment, we compare DINOv2, MAE, CLIP [24], and SigLIP [31] against the 11 language models for a total of $4 \times 11 = 44$ text–vision model pairs. All four are ViT-Base encoders [4] with different pretraining objectives. Alignment is computed as in Section 2 over 1,024 image–caption pairs from the Wikipedia-based Image Text (WIT) dataset [26].

For tractability, we compute mean alignment over a subset of the layers. On the text side, we consider at most[4] five evenly spaced layers from the band $\mathcal{B}_A$. On the vision side, we consider the final hidden state along with five evenly spaced blocks from index 2 to $\mathrm{final} - 2$. For a model pair, this gives at most $5 \times 6 = 30$ layer pairs.

We consider a competence axis only on the text side, using the same HellaSwag benchmark.

3.3 Text–text shared J-lens output

For our third experiment, we apply the prefitted J-lenses per model and per layer over 200 documents in the pile-10k dataset [19]. Each document is first truncated to at most 1,500 characters, then to at most 300 tokens. We read the lens out as words, keeping the top $t = 25$ entries.

$$\mathrm{lens}_\ell^A(h) \;=\; \mathrm{softmax}\big(W_U^A \, \mathrm{norm}^A\big(J_\ell^A h\big)\big)\,.$$

Since the 11 models do not share a tokeniser, we restrict the J-lens to the 31,548 strings that are a single token in all 11 vocabularies. We also index positions by character rather than by token, evaluating at 6,000 character positions that fall at a token-end word boundary in all 11 tokenisers.

We then measure how many of the $t = 25$ words are shared between model pairs as a function of relative layer depth $\lambda = \ell/(L_A - 1)$. We compare against the logit lens [20], which outputs the next-token distribution if stopped at a given layer by reading the residual stream directly: $\mathrm{softmax}(W_U^A \, \mathrm{norm}^A(h))$.

We then relate J-lens overlap to mean HellaSwag accuracy for each model pair to re-evaluate the convergence trend measured in Experiment 1.

[3] All benchmark results were self-computed to ensure consistency.

[4] The layer band for Pythia-70M contains only two layers. All other text models use five layers.

Table 1: Alignment on real versus content-ablated documents over 55 model pairs. Retention is the mean over all pairs of content-ablated alignment divided by the original alignment, and is substantially lower for the J-space than for the full space. The lower block repeats the ablation with the J-lens dictionary replaced by a Gaussian dictionary. Bracketed intervals are 95% percentile bootstrap confidence intervals over the 55 model pairs (2,000 resamples).

	Real alignment	Content-ablated alignment	Retention
Components, J-lens dictionary			
full	0.5577 [0.539, 0.579]	0.3009 [0.290, 0.312]	54.1% [53.01, 55.14]
non-J	0.5057 [0.484, 0.529]	0.2609 [0.252, 0.271]	52.0% [50.73, 53.30]
J	0.4202 [0.409, 0.432]	0.1099 [0.108, 0.112]	**26.3%** [25.63, 27.02]
J-space under a substituted dictionary			
Gaussian	0.2103 [0.202, 0.220]	0.0719 [0.071, 0.073]	34.7% [33.66, 35.85]

All experiments were run on a single RTX 3080 (10 GB) GPU, taking a total of around 40 hours.

4 Results

4.1 Text–text alignment and motivating experiment

When we randomise all content words, the full alignment retains a mean of 54.1% of its original value, while J-space alignment retains only 26.3% (Table 1). To test how much of this drop comes from the sparse coding itself, we repeat the ablation experiment with a randomised Gaussian dictionary instead of the J-space dictionary. Under this control, retention is 34.7%, meaning sparse coding accounts for 69.7% of the drop between full and J-space retention. However, the remaining third is still attributable to the J-space itself.

J-space alignment therefore depends more heavily on content than the full activation space.

Mean alignment for the J-space (0.420), the non-J remainder (0.506), and the full activation space (0.558) is between $42.0\times$ and $55.7\times$ the chance level.

For each model pair, we compare alignment and the mean of their competences. We find a positive correlation across the J-space (Spearman correlation $\rho = +0.56$, $p = 0.0162$), full activation ($\rho = +0.77, p = 0.0008$), and remainder ($\rho = +0.87, p < 0.0001$) (Figure 2(a)). Since pairs are not independent, p-values come from permuting competence across models while holding alignment values fixed.

The rate of convergence in the J-space is about a third of that in the full activation space. Alignment rises with pair competence at $+0.76 \pm 0.09$ per unit HellaSwag accuracy in the full activation space against $+0.23 \pm 0.07$ in the J-space.

4.2 Text–vision alignment

Alignment between text models and vision encoders for each component is between $6.8\times$ and $7.4\times$ the chance level. It correlates strongly and positively with competence across the J-space ($\rho = +0.97$, $p < 10^{-4}$), the full activation space ($\rho = +0.93$, $p = 0.0001$), and the remainder ($\rho = +0.92$, $p = 0.0002$) (Figure 2(b)).

Unlike the text–text comparison, we do not correlate competence with alignment across all 44 text–vision pairs—since the four encoders differ from one another far more than the text models do, this would obscure the actual trend. Indeed, mean J-space alignment ranges from 0.053 for MAE to 0.087 for SigLIP, a spread well in excess of the standard deviation of cross-modal alignment across the 11 text models (0.0134).

We instead average alignment within each encoder, then average these four values. We report the four within-encoder J-space correlations, which range from $+0.95$ to $+0.97$, in Appendix A.

Cross-modally, there is far less shared superficial structure for a full-space comparison to capture, so intuitively, J-space and full alignment should match, and they do. This is further evidence that the

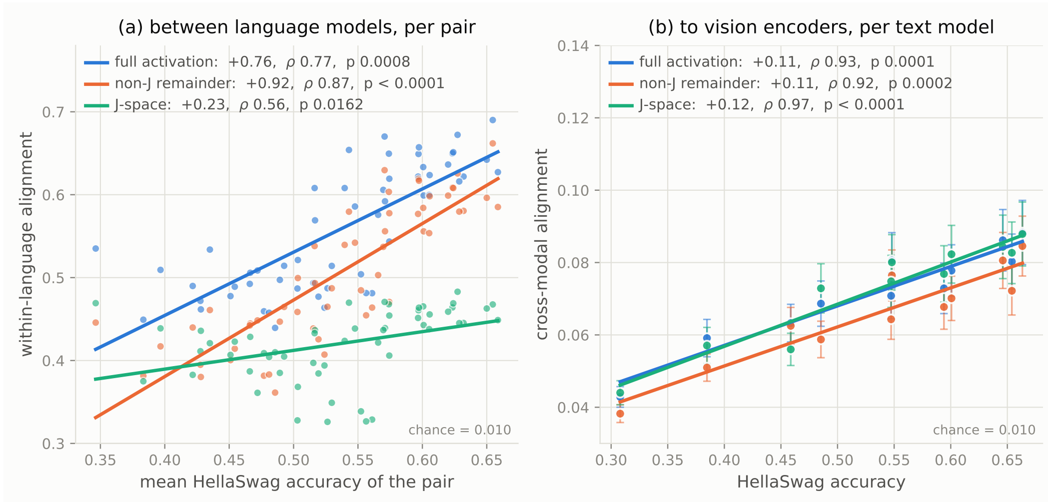


Figure 2: Alignment graphed against HellaSwag accuracy for full activation, J-space, and non-J remainder, with a line of best fit. **(a)** Text–text alignment against competence; p-values are model-label permutation values. The J-space converges much more slowly than the full activation space. **(b)** Alignment averaged over the four vision encoders; error bars show standard error of this mean across the four encoders. Alignment rises with competence for every component in both settings.

gap between these components within language is explained by superficial structure rather than by a property of the J-space.

Results for other metrics are reported in Appendix D.

4.3 Text–text shared J-lens output

We find that at matched relative layer depth, about 5.1 of 25 words are shared, against 2.0 of 25 for the plain logit lens. In particular, for middle layers $\lambda \in [0.2, 0.5]$, J-lens alignment beats logit lens alignment by 4–5$\times$. The logit lens comparison rules out the explanation that J-lens alignment arises because models might agree on next-token predictions.

We also find that across the 38 cross-family model pairs, a model's best-matching counterpart layer sits, on average, 11.1% of the layer stack away, consistent with Figure 3.

Finally, J-lens overlap rises with competence, further validating the results of Section 4.1. Over the same 55 pairs, overlap and competence correlate at $\rho = +0.74$ ($p = 0.0020$), shown in Figure 4. The plain logit lens gives $\rho = +0.41$, which does not reach significance ($p = 0.11$).

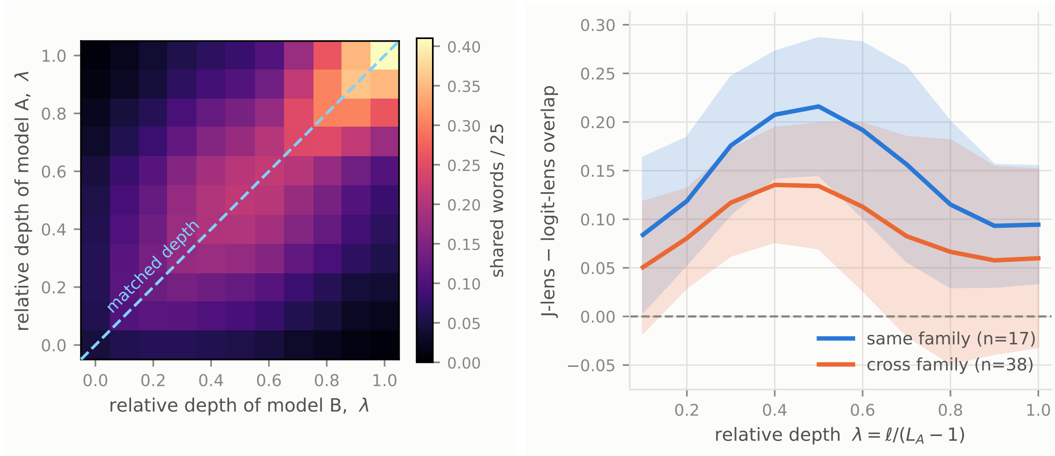
This readable form of alignment correlates with competence at least as strongly as the kernel form, corroborating the convergence claim.

4.4 Controls

In addition to the random-dictionary control above, we run five further controls:

Random-unembedding-row null When we replace the J-lens dictionary with random unembedding rows, J-space alignment exceeds the strongest of five seeded draws in all 55 pairs (mean margin $+0.167$). This shows that J-space alignment captures information that random vocabulary directions do not. The null dictionaries use a random subset of 20,000 unembedding rows rather than the full dictionary due to computational constraints.

Lens-fitting corpus To rule out the possibility that J-space alignment emerges from the shared fitting corpus, we refit so that no two compared models share one. The effect is immaterial (Appendix C.1).



(a) **(b)**

Figure 3: **(a)** Mean J-lens top-25 word agreement between language models as a function of each model's relative layer depth $\lambda = \ell/(L_A - 1)$. Grids are symmetrised before averaging. **(b)** The difference between J-lens and logit lens overlap, against relative layer depth. $\lambda = 0$ is omitted as the logit lens reads the embedding matrix directly, and so dominates by construction. Lines are shown for same-family and cross-family pairs. The shaded region is ± 1 standard deviation across the model pairs in each family group. The mean difference is positive at every depth shown, and peaks in the middle layers.

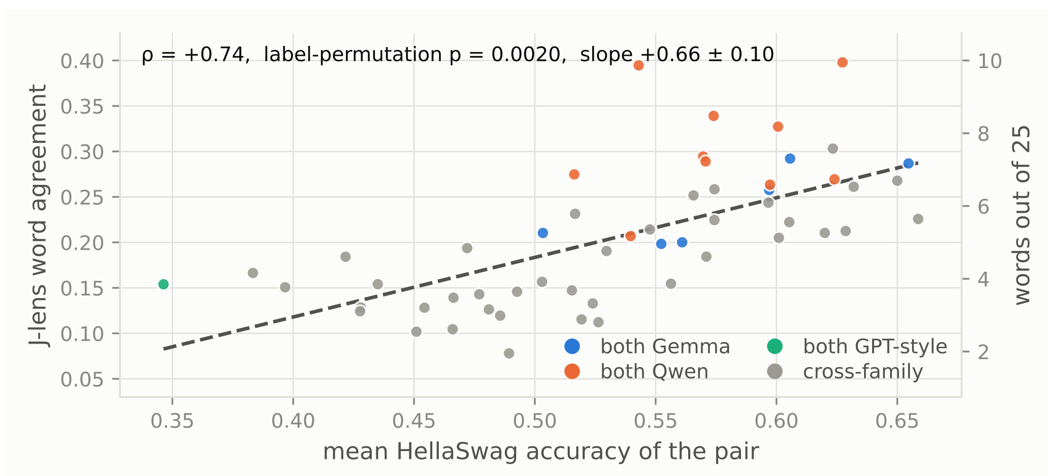


Figure 4: J-lens top-25 median overlap over all matched relative layer depths, against the pair's mean HellaSwag accuracy. Points are coloured by family composition, and p is a model-label permutation value. Agreement rises with competence.

Shuffling image–caption pairs Shuffling image–caption pairings in the text–vision alignment experiments reduces alignment to an average of 0.0096, against a chance level of 0.0098. Since shuffled pairs share no image–caption relationships, the alignment we measure on the actual pairings is not simply a result of the metric.

Lens degradation The J-lens is fitted on ordinary text, so the reduced alignment on the content-ablated documents in the first experiment may result from the J-lens simply working less well on such documents. We find that degrading the lens cannot reproduce the effect of removing semantic meaning (Appendix C.2).

Position-shuffled floor In Experiment 3, some of the top-25 lens overlap is coincidental, since both models naturally rank common tokens highly. We compare each model's top-25 list against the other model's list at a randomly chosen position in the corpus, which preserves the marginal distributions over words while destroying the correspondence between them. Under this control, overlap is 0.1 of 25 words for both the J-lens and logit lens, which is negligible.

5 Conclusion and limitations

We propose measuring Platonic convergence in the J-space to isolate shared semantic content. Our results suggest that full-space convergence is inflated by surface-level agreement. Within language (where models share superficial regularities), restricting the comparison to the J-space slows convergence to a third of the full-space rate. Across modalities, where no such regularities exist, the J-space and full components converge at rates that are indistinguishable within standard error. Nevertheless, the original convergence claimed by the Platonic Representation Hypothesis still stands, as the J-spaces and J-lenses still align more at higher competence.

This alignment is meaningful and can be read out as actual words. At the same relative layer depth, language models' J-lenses have significant overlap, and this overlap is greater than with the plain logit lens. Measuring convergence with the J-lens reproduces the result of the first experiment, finding that more competent models share a greater J-lens overlap.

Limitations First, we use limited compute for experiments, testing only 11 text models and four vision encoders. The text models span 70M–4.3B parameters, and all four vision encoders are ViT-Base, so we have no competence axis on the vision side. HellaSwag is near its floor for our smallest models: Pythia-70M scores 30.8% and GPT-2 scores 38.5% against 25% chance. Thus, competence differences at the bottom of the range are highly compressed. We also compute alignment only on about 1,000 documents, a scale that Koepke et al. [14] argue inflates alignment, though our claim is relative.

Second, we show the J-space's sensitivity to semantic content through only the retention statistic and a single random-dictionary control.

Third, our cross-modal measurement is asymmetric, as for a given image–caption pair, we compare the J-space of a text model to the full visual representation of the vision encoder.

Fourth, the J-space is a small part of the full activation space, accounting for an average of 18.3% of the pooled activation's squared norm across the 11 models. The non-J remainder therefore carries the other 81.7% of the squared norm. Our results concern a minority component of the full activation space, albeit a privileged one, and the remainder is close enough to the full activation that the two behave similarly.

Fifth, our main results use the mutual κ-nearest-neighbour measure of kernel alignment. Appendix D repeats the first two experiments under seven new metrics, keeping all else the same. The claim about within-language J-space convergence only holds under neighbourhood-based measures, but not global subspace measures; cross-modal results hold under all measures. In the appendix we explain this result and justify why we believe global subspace metrics are not appropriate measures of alignment in the J-space.

Finally, the J-lens is fitted per token position, but we apply it to mean-pooled document vectors. This means we are assuming that the dictionary's validity transfers to those vectors.

Next steps We would like to strengthen our key claim about the J-space's semantic sensitivity with more tests on larger and more varied model types.

We believe that by leveraging the semantically selective properties of the J-space, we can re-examine many other questions in mechanistic interpretability. We would also welcome a theoretical explanation for our findings.

Finally, our third experiment compares word overlap rather than activation geometry, a comparison that is more functional than representational. Since this paper mostly compares representations, we would welcome future research that uses the J-space to measure functional convergence.

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A Aggregation robustness

Table 2 gives the cross-modal correlation separately for each vision encoder. Each encoder has a different alignment level, which is why we consider correlations within encoders rather than pooling across all 44 pairs. The relationship holds within every encoder, including MAE, which receives no text supervision at all.

Table 2: Cross-modal alignment against HellaSwag accuracy, computed separately within each vision encoder over the 11 text models. The "mean J" column shows alignment for a given encoder across the 11 text models, highlighting the differences between encoders that make pooling the 44 pairs inadvisable. p-values are exact permutation values for $n = 11$.

	mean J	full	non-J	J
DINOv2	0.0742	$\rho = +0.93\ {\scriptstyle p\,=\,0.0001}$	$\rho = +0.92\ {\scriptstyle p\,=\,0.0002}$	$\rho = +0.97\ {\scriptstyle p\,<\,0.0001}$
MAE	0.0535	$\rho = +0.91\ {\scriptstyle p\,=\,0.0003}$	$\rho = +0.92\ {\scriptstyle p\,=\,0.0002}$	$\rho = +0.95\ {\scriptstyle p\,<\,0.0001}$
CLIP	0.0764	$\rho = +0.93\ {\scriptstyle p\,=\,0.0001}$	$\rho = +0.92\ {\scriptstyle p\,=\,0.0002}$	$\rho = +0.97\ {\scriptstyle p\,<\,0.0001}$
SigLIP	0.0865	$\rho = +0.93\ {\scriptstyle p\,=\,0.0001}$	$\rho = +0.92\ {\scriptstyle p\,=\,0.0002}$	$\rho = +0.97\ {\scriptstyle p\,<\,0.0001}$

B Competence axes

The main paper measures competence with HellaSwag exclusively. Holding alignment values fixed, we vary the competence axis and derive similar results, shown in Table 3 and Table 4.

Table 3: Spearman correlation between within-language alignment and mean competence of the two models, per component. We use both HellaSwag and log parameter count as competence axes. The pairs are not independent, so we use model-label permutation p-values, which hold the alignment values fixed and permute competence across models. The alignment values are the same as in Table 1, and only the competence axis varies.

	log parameters ($n = 55$)	HellaSwag ($n = 55$)
full	$\rho = +0.79\ {\scriptstyle p\,=\,0.0006}$	$\rho = +0.77\ {\scriptstyle p\,=\,0.0008}$
non-J	$\rho = +0.87\ {\scriptstyle p\,=\,0.0002}$	$\rho = +0.87\ {\scriptstyle p\,<\,0.0001}$
J	$\rho = +0.60\ {\scriptstyle p\,=\,0.0094}$	$\rho = +0.56\ {\scriptstyle p\,=\,0.0162}$

Table 4 repeats the cross-modal comparison of Figure 2(b) against all three competence measures. We also measured competence using language-modelling performance, computed as $1 - \text{bits-per-byte}$ over 4M tokens of OpenWebText [9]. We derive similar results to those in the main paper.

We follow Huh et al. [13] in using this competence measure for the cross-modal comparison only; we note that it gives a within-language J-space correlation of $\rho = +0.35$ ($p = 0.185$), which is not significant.

Table 4: Spearman rank correlation between cross-modal alignment and competence, per component. Each of the 11 text models contributes one point, its alignment averaged over the four vision encoders and each layer-pair grid reduced by its mean; p-values are exact permutation values for $n = 11$. As in Table 3, only the competence axis varies, here log parameter count, HellaSwag, and $1 - \text{bits-per-byte}$.

	log parameters	HellaSwag	$1 - \text{bits-per-byte}$
full	$\rho = +0.84\ {\scriptstyle p\,=\,0.0018}$	$\rho = +0.93\ {\scriptstyle p\,=\,0.0001}$	$\rho = +0.96\ {\scriptstyle p\,<\,0.0001}$
non-J	$\rho = +0.83\ {\scriptstyle p\,=\,0.0022}$	$\rho = +0.92\ {\scriptstyle p\,=\,0.0002}$	$\rho = +0.98\ {\scriptstyle p\,<\,0.0001}$
J	$\rho = +0.89\ {\scriptstyle p\,=\,0.0005}$	$\rho = +0.97\ {\scriptstyle p\,<\,0.0001}$	$\rho = +0.94\ {\scriptstyle p\,<\,0.0001}$

C Lens robustness controls

C.1 Lens-fitting control

In the main experiment, J-lenses were all fitted on the same corpus (WikiText-103 Merity et al. [18]). As a result, J-space alignment might be a result of the fact that the lenses were fitted on shared text. Figure 5 recomputes text–text alignment, splitting the fitting corpus into two halves and ensuring the J-lenses of two models in a pair do not use the same half. We also repeat the experiment where two models in a pair share the same half of the fitting corpus. Due to computational constraints, we conduct the control on only 10 pairs (spanning three families) among five models: Pythia-70M, GPT-2, Gemma-3-270M, Qwen3.5-0.8B, and Gemma-3-1B.

We find that corpus-sharing has no measurable impact on alignment. Per-pair differences between the shared-half and crossed-half conditions have a median of $+0.0002$. All three alignment conditions remain far above the random-dictionary control for every pair.

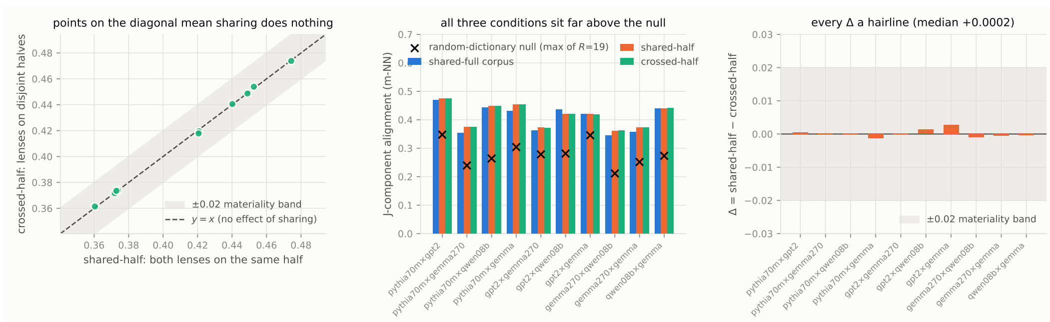

Figure 5: Disjoint lens-fitting control over the 10 pairs among five models. Lenses are fitted on disjoint halves of WikiText-103. **Left:** crossed-half against shared-half J-component alignment; points lie on $y = x$ within the ± 0.02 materiality band fixed in advance, so refitting on disjoint halves changes nothing. **Middle:** the three fitting conditions per pair—shared-full corpus, shared-half, and crossed-half—all sit far above the per-pair random-dictionary control. **Right:** the per-pair difference Δ between shared-half and crossed-half, median $+0.0002$ against an alignment level of 0.42, every value inside the band.

C.2 Lens-degradation control

The J-lens was fitted only on ordinary text, not content-ablated text, so the reduced alignment on the content-ablated documents may be a result of the J-lens simply working less well on such documents. Indeed, the J-space component accounts for 13.5% of the activation's squared norm on content-ablated text against 18.3% on real text.

We construct an instrument-only null by reducing the number of dictionary directions on the original corpus until the J-space component accounts for the same fraction of the activation's squared norm as it does on content-ablated text. This gives a J-space that is degraded but computed on documents with intact meaning. Alignment between model pairs with the degraded J-space is 0.2924 against 0.1099 for the content-ablated text. Even a five-direction J-space, which accounts for less of the activation than the 25-direction J-space on content-ablated text, gives alignment $2.1\times$ higher (0.2326 against 0.1099).

Thus, lens degradation cannot reproduce the effect of removing semantic meaning.

D Varying the alignment metric

The main text measures alignment through mutual κ-nearest-neighbour overlap. We vary our choice of metric, changing only the function for alignment between two layer pairs (Table 5). Implementations follow Huh et al. [13].

We note that within language, regardless of metric, full alignment is always greater than J-space alignment. Cross-modally the two are near-indistinguishable, with J-space marginally ahead for six of the eight metrics, while the cross-modal correlation result is similarly strong ($+0.94 \leq \rho \leq +0.97$).

Within-language, metrics split into two groups. Metrics which compare *local neighbourhood structure* (e.g. κ-NN) follow our main paper's results, and metrics which compare *global subspace geometry*, i.e. how variance is distributed, do not. Furthermore, the competence–alignment correlations for neighbourhood-based metrics all reach significance with $p < 0.02$, but the correlations for global subspace metrics are not significant (the smallest is $p = 0.267$).

Global subspace metrics are insensitive to semantic content We repeat the content-ablation experiment of Section 4.1, again only varying the metric. Results are shown in Table 6. Under the three global metrics, we find that full-space alignment is essentially unchanged by the ablation: each

Table 5: J-space and full alignment on different metrics, with all other parts of the experiment identical to the main paper. $\rho(\mathrm{J})$ is the J-space correlation between alignment and mean HellaSwag competence, J/full is the ratio of the rates of convergence. p-values are model-label permutation values. Neighbourhood metrics reproduce main results, while global metrics do not (see the paragraph below).

	Within language ($n = 55$)				Cross-modal ($n = 11$)		
Metric	full	J	ρ(J)	J/full slope	full	J	ρ(J)
Neighbourhood-based							
mutual κ-NN (ours)	0.5577	0.4202	$+0.56$ $p = 0.016$	$+0.30$	0.0719	0.0726	$+0.97$ $p < 0.0001$
CKNNA	0.5975	0.4615	$+0.63$ $p = 0.011$	$+0.43$	0.0761	0.0763	$+0.96$ $p < 0.0001$
cycle κ-NN	0.9358	0.9122	$+0.63$ $p = 0.011$	$+0.68$	0.3672	0.3712	$+0.95$ $p < 0.0001$
edit-distance κ-NN	0.1609	0.0936	$+0.62$ $p = 0.011$	$+0.23$	0.0106	0.0106	$+0.95$ $p < 0.0001$
LCS κ-NN	3.5457	2.6809	$+0.60$ $p = 0.012$	$+0.28$	0.5712	0.5762	$+0.97$ $p < 0.0001$
Global subspace							
CKA	0.8036	0.6892	$+0.09$ $p = 0.651$	-0.00	0.2001	0.2038	$+0.97$ $p < 0.0001$
unbiased CKA	0.7938	0.6615	-0.19 $p = 0.475$	-0.31	0.1687	0.1677	$+0.94$ $p = 0.0001$
SVCCA	0.6846	0.5869	-0.30 $p = 0.267$	-0.55	0.3070	0.3029	$+0.94$ $p < 0.0001$

metric retains at least 96.9%. Notably, ablating semantic content nominally increases these metrics' alignment as shown by CKA and unbiased CKA both measuring retention values above 100%. For J-space ablation, the same metrics retain 88.1%, 85.4%, and 77.6%. In other words, under a quarter of J-space alignment is attributable to semantic content. For neighbourhood-based measures, the opposite is true. Thus, since this paper seeks to measure semantically significant convergence, global subspace metrics are not appropriate.

Furthermore, we argue that such metrics are not an accurate measure of J-space alignment, since we compute the J-space as a sparse combination of only 25 vocabulary directions. Thus, its coarse geometry is quite generic, and every model's J-space points in broadly similar directions. As a result, global measures see all J-spaces as similar. For example, GPT-2 and Pythia-70M, our two smallest models, have the highest J-space alignment under CKA.

Since neighbourhood-based metrics and global metrics are two ends of a spectrum, we can find at what locality the correlations begin to decay. CKNNA restricts the kernel to an item's k nearest neighbours before computing CKA, so for $k = n - 1$, it is simply unbiased CKA. As we vary k across the spectrum, we find that J-space correlation decays monotonically, from $\rho = +0.68$ at $k = 2$ to $\rho = -0.19$ at $k = 999$. The correlation loses significance for k around 50–100. Thus, the within-language claim in our main paper holds for neighbourhoods up to roughly 5% of the corpus.

The percentage of the drop in retention explained by the J-space matches our main results only for the neighbour-identity metrics (mutual κ-NN, LCS, and edit-distance). We note that the global subspace metrics carry little weight here, as they capture very limited semantic content.

Table 6: Retention after the content-ablation experiment, by metric. Retention for the random-Gaussian dictionary is also shown, as well as the percent drop explained by the J-space: $(\text{Gaussian} - \text{J})/(\text{full} - \text{J})$.

	Retention			
Metric	full	J	Gaussian	Explained by J-space
Neighbourhood-based				
mutual κ-NN (ours)	54.1%	26.3%	34.7%	30.3%
CKNNA	49.8%	20.2%	19.3%	−3.0%
cycle κ-NN	64.4%	45.0%	44.7%	−1.8%
edit-distance κ-NN	37.7%	18.5%	27.7%	47.9%
LCS κ-NN	57.0%	31.5%	39.3%	30.6%
Global subspace				
CKA	100.8%	88.1%	82.7%	−42.1%
unbiased CKA	101.4%	85.4%	117.9%	203.5%
SVCCA	96.9%	77.6%	79.2%	8.7%

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